

The pad part of the name of this encryption technique comes from the fact that early implementations involved a pad of paper where the top sheet could be torn off and destroyed after use. Subsequently, huge reductions in the size of the pad was possible. KGB went ahead and created pads that fit in the palm of the hand, or even in a walnut shell. In fact, in a very mission impossible-ish turn of events, highly inflammable sheets of nitrocellulose were once used as one-time pads.

Despite being brilliant in theory, there are certain practical issues with the usage of one-time pads. First and foremost, perfect randomness is required in the one-time pads, which can't exactly be obtained from run of the mill software. High-quality random number generation is difficult to achieve. The random number generation function in most programming languages is not fit for cryptographic use. True randomness is expected to the extent that even if the generating process of a sequence were known up to the last point, it would still be impossible to predict the next event. This simply rules out most running software on computers, which use deterministic processes for random number generation. The next event in a computer is actually pretty predictable, hence they're unusable.

Radioactive decay is an example of a non-deterministic event. Secure generation and exchange of one-time pad material, which must be as long as the plaintext, is another headache. Most importantly, ensuring its proper disposal to avoid reuse and keeping it secret from any adversary is also an issue.

The pad must be kept secure and has to be as long as (or longer than) the message. Also, you wouldn't want to send short messages with a one-time pad, and thus a secure long pad can be used to send numerous messages until you run out of that particular pad cipher.

These practical considerations introduce vulnerabilities which hamper real-world application. Nowadays, one-time pads are rarely used. Implementation difficulties have led to the one-time pad being broken time and again, with breaches serious enough to discourage its adoption as a wide-spread information security tool.

Since the pad is too long for any human to remember, you have to carry them in storage media such as pen drives, DVDs, etc. This is cumbersome, especially when you compare it to modern public key crypto-systems. The risk of compromise during the transportation of the pad, such as someone stealing, copying and returning the pad also cannot be ignored. The effort needed to manage one-time pads for large networks is also a scalability issue.